

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: _____Zhang Ka Chun_____

Student ID: _____24023973d_____

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

_____show ip route ospf_____

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? __S0/0/0__

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? __129__

Does R2 show up as an OSPF neighbor on R1? __Yes__

Does R2 show up as an OSPF neighbor on R3? __No__

What does this information tell you?

Traffic destined for 192.168.2.0/24 from R3 is taking the path through R1. This happens because R2's S0/0/1 interface remains in passive mode, preventing it from sending OSPF updates to R3.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_____R2(config)# router ospf 1_____

_____R2(config-router)# no passive-interface s0/0/1_____

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? __S0/0/1__

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65. The total is the sum of the T1 serial link's cost (64) and the Gigabit LAN interface's cost (1) on R2.

Is R2 listed as an OSPF neighbor to R3? __Yes__

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Since modern network interfaces often operate at speeds exceeding 100 Mbps, increasing the reference bandwidth ensures that OSPF can accurately differentiate and calculate costs for these high-speed links.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

For the 192.168.3.0/24 network, the metric is 782 (R1's S0/0/1 cost of 781 plus R3's G0/0 cost of 1).
For 192.168.23.0/30, the metric is 845 (R1's S0/0/1 cost of 781 plus R3's S0/0/1 cost of 64).

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1,562. Because the route to 192.168.23.0/24 passes through two serial connections, and each now has a cost of 781, the total accumulated cost is $781 + 781 = 1,562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Since OSPF always prefers the lowest-cost path, it selects the route via R2 (total cost 1,563: $781+781+1$) instead of the direct link to R3, which now has a higher cost of 1,566 ($1565+1$).

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The Router ID is crucial for the DR and BDR election in multi-access networks. Using a loopback address or static ID prevents the router ID from changing if a physical interface fails, thereby maintaining network stability.

2. Why is the DR/BDR election process not a concern in this lab?

DR and BDR elections only occur on broadcast multi-access networks (like Ethernet). Because this topology uses point-to-point serial connections between routers, the election process is bypassed.

3. Why would you want to set an OSPF interface to passive?

Setting an interface to passive stops OSPF from sending hello packets and routing updates out that port, which saves bandwidth and increases security on LANs, while still allowing the router to advertise that network to other OSPF neighbors.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF utilizes a multi-area design (centering around a backbone Area 0), which establishes a hierarchical structure. This limits the scope of routing updates and allows the network to scale efficiently.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The router will select the EIGRP route. This is determined by Administrative Distance (AD), which ranks the trustworthiness of routing protocols. EIGRP's AD of 90 is lower (more trusted) than OSPF's 110 and RIP's 120, so metric values are not compared across different protocols.